

- Grötzsch graph (*grotzsch_graph()*) – A triangle-free graph with 11 vertices and 20 edges, named after Herbert Grötzsch.
- Heawood graph (*heawood_graph()*) – A symmetrical graph with 14 vertices and 21 edges, named after Percy John Heawood.
- Petersen graph (*petersen_graph()*) – A symmetrical graph with 10 vertices and 15 edges, named after Julius Petersen.
- Tutte graph (*tutte_graph()*) – A graph with 46 vertices and 69 edges, such that every vertex has three neighbours. It is named after W T Tutte.

And here follows a demonstration of some of the above, along with the visuals (left to right, top to bottom) in Figure 5:

```
$ maxima -q
(%i1) load(graphs)$
...
@ errors, @ warnings
(%i2) g: circulant_graph(10, [1, 3]);
(%o2)          GRAPH(10 vertices, 20 edges)
(%i3) draw_graph(g, show_id=true, vertex_color=yellow)$
(%i4) g: circulant_graph(10, [1, 3, 4, 6]);
(%o4)          GRAPH(10 vertices, 40 edges)
(%i5) draw_graph(g, show_id=true, vertex_color=yellow)$
(%i6) g: flower_snark(3);
(%o6)          GRAPH(12 vertices, 18 edges)
(%i7) draw_graph(g, show_id=true, vertex_color=yellow)$
(%i8) g: flower_snark(5);
(%o8)          GRAPH(20 vertices, 30 edges)
(%i9) draw_graph(g, show_id=true, vertex_color=yellow)$
(%i10) g: flower_snark(8);
(%o10)         GRAPH(32 vertices, 48 edges)
(%i11) draw_graph(g, show_id=true, vertex_color=yellow)$
(%i12) g: flower_snark(10);
(%o12)         GRAPH(40 vertices, 60 edges)
(%i13) draw_graph(g, show_id=true, vertex_color=yellow)$
(%i14) g: wheel_graph(3);
(%o14)         GRAPH(4 vertices, 6 edges)
(%i15) draw_graph(g, show_id=true, vertex_color=yellow)$
(%i16) g: wheel_graph(4);
(%o16)         GRAPH(5 vertices, 8 edges)
(%i17) draw_graph(g, show_id=true, vertex_color=yellow)$
(%i18) g: wheel_graph(5);
(%o18)         GRAPH(6 vertices, 10 edges)
(%i19) draw_graph(g, show_id=true, vertex_color=yellow)$
(%i20) g: wheel_graph(10);
(%o20)         GRAPH(11 vertices, 20 edges)
(%i21) draw_graph(g, show_id=true, vertex_color=yellow)$
(%i22) g: wheel_graph(100);
(%o22)         GRAPH(101 vertices, 200 edges)
(%i23) draw_graph(g, show_id=true, vertex_color=yellow)$
(%i24) g: clebsch_graph();
(%o24)         GRAPH(16 vertices, 40 edges)
(%i25) draw_graph(g, show_id=true, vertex_color=yellow)$
(%i26) g: grotzsch_graph();
```

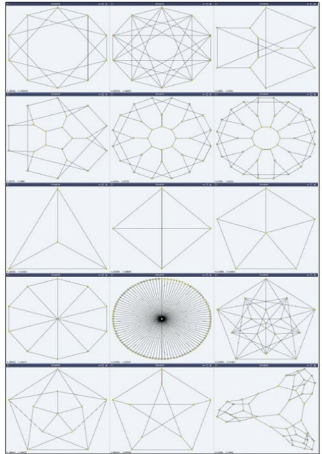


Figure 5: Graph beauties

```
(%o26)          GRAPH(11 vertices, 20 edges)
(%i27) draw_graph(g, show_id=true, vertex_color=yellow)$
(%i28) g: petersen_graph();
(%o28)          GRAPH(10 vertices, 15 edges)
(%i29) draw_graph(g, show_id=true, vertex_color=yellow)$
(%i30) g: tutte_graph();
(%o30)          GRAPH(46 vertices, 69 edges)
(%i31) draw_graph(g, show_id=true, vertex_color=yellow)$
(%i32) quit();
```

What next?

You may be wondering what to do with all these visualisations, apart from just staring at them. Visualisations were just to motivate you. In fact, every graph has a particular set of properties, which distinguishes it from the other. And, there is a lot of beautiful mathematics involved in these properties. If you are motivated enough, try playing around with these graphs and their properties. 

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